5. Induction

The one idea the whole machine rests on

A *changing* magnetic field makes a voltage

Lesson 5 · about 90 minutes · everyone builds; no external power at all

What you need

- Magnet wire, 26–30 AWG, 30 m or more
- A cardboard tube (kitchen roll) and a length of copper pipe
- Strong neodymium magnets that drop freely through both
- Multimeter with a millivolt range
- Two LEDs wired back-to-back
- An aluminium tube, if you have one

Predict first

You drop a magnet down a copper pipe. Copper is not magnetic and the magnet never touches the walls. Does it fall faster than, slower than, or the same as a magnet dropped down a cardboard tube of the same size? Commit before you drop anything.

Do it

1. **Wind a sense coil.** Wind 200–400 turns of magnet wire around the cardboard tube. Sand the ends and connect them to the multimeter on its millivolt DC range.
2. **Drop the magnet.** Drop it through. The meter jumps and returns to zero.
3. **Hold it still.** Rest the magnet inside the coil and leave it there. The reading is **zero**. A magnetic field alone produces nothing.
4. **Change one thing at a time.** Drop it faster; use two magnets stuck together; turn the magnet over. Record the size and the *sign* each time.
5. **Light an LED.** Replace the meter with two LEDs wired in opposite directions across the coil ends. Move the magnet in and out briskly. One LED flashes going in, the other coming out.
6. **The copper pipe.** Now drop a magnet down the cardboard tube and down the copper pipe, and time them. Compare with your prediction. Try the aluminium tube too.

What it means

Faraday's law	A voltage appears in a loop when the magnetic field *through* that loop changes. Not when it is strong — when it *changes*. Faster change, bigger voltage; more turns, bigger voltage.
Why holding it still gives nothing	This is the whole lesson. A stationary magnet inside a coil produces exactly zero. Every generator, every transformer and every Tesla coil exists to make something change.
Lenz's law	The two LEDs flash on opposite motions because the induced current always opposes the change that made it. Push a magnet in and the coil pushes back.
The copper pipe	The falling magnet induces circulating eddy currents in the pipe wall. Those currents make their own magnetic field, which by Lenz's law opposes the fall. The magnet falls in slow motion. Nothing touches it and copper is not magnetic — it is being braked by a field that its own falling created. This is the most surprising thing in the course, and it is worth doing several times.
Conservation	The energy in the flash came from your arm. Moving a magnet into a coil is *harder* when the circuit is closed than when it is open. Try it with the LEDs connected and disconnected and feel the difference.

The link to the coil — read this one twice

A Tesla coil's secondary is a coil sitting inside the primary's magnetic field. When the spark gap fires, the primary's current swings from maximum to zero and back hundreds of thousands of times a second. That violently changing field induces a voltage in the secondary, and because the secondary has roughly a thousand times more turns, and because both circuits are tuned to the same frequency so the effect builds up over many cycles, the result is hundreds of thousands of volts. Lessons 6, 8 and 9 each add one clause to that sentence. Every one of them depends on what you just measured: *change*, not strength.

> **This lesson is low voltage.** Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.